WEEK 2

Week 2: Cluster Preparation with Proxmox - Lab Guide

HPC & Cluster Computing

Objective: Build your HPC environment from scratch on the VUT Proxmox Cluster infrastructure.

Goal: Access Proxmox, create and configure your Master Node (including networking, SSH keys, and environment variables), and then create empty Compute Node VMs to prepare for network provisioning in Week 3.

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1. Accessing Proxmox & Uploading ISO

Proxmox VE is an enterprise-grade virtualization management platform. You will build your cluster VMs here.

1. Log into the Proxmox Web Interface (URL and credentials provided by the instructor).
2. Download the openEuler ISO.
3. In the left pane of Proxmox, select your designated storage (e.g., **local**).
4. Click **ISO Images** > **Upload** and upload the openEuler ISO.

2. Creating the Master Node

1. Click **Create VM**. Set the Name to `hpc-master-node`.
2. **OS:** Select the uploaded openEuler ISO.
3. **Disks:** Allocate at least 20 GB.
4. **CPU & Memory:** Allocate 2 Cores and 2048 MB (2 GB) RAM.
5. **Network:** Leave default (e.g., `vmbr0` for internet). Finish VM creation.
6. Select `hpc-master-node`, go to **Hardware** > **Add Network Device**. Select the internal cluster bridge (e.g., `vmbr1`). **Repeat this step** so the master node has two internal network devices (these will become `eth1` and `eth2`).
7. Start the VM, open **Console**, and install openEuler (setup root password, create user).

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3. Master Node Configuration

Log into your newly installed openEuler Master Node.

3.1 Hostname and Static IP

We are setting the system hostname to "master" and creating a bonded network interface (`bond0`) using both internal NICs (`eth1` and `eth2`) for maximum throughput and redundancy. We assign it the static IP `10.0.0.1`.

```
hostnamectl set-hostname master
# Create the bond interface
nmcli connection add type bond con-name bond0 ifname bond
# Attach the physical interfaces (eth1, eth2) as slaves t
nmcli connection add type ethernet slave-type bond con-na
nmcli connection add type ethernet slave-type bond con-na
```

```
# Activate the bond
nmcli connection up bond0
```

3.2 SSH Keys & Passwordless Login

Generate SSH keys and immediately authorize them. Because we are doing this *before* cloning, any clones of this VM will inherently trust the Master, enabling instant passwordless SSH across the cluster!

```
ssh-keygen -t ed25519 -f ~/.ssh/id_ed25519 -N ""
cat ~/.ssh/id_ed25519.pub >> ~/.ssh/authorized_keys
```

3.3 Cluster Environment Variables

Consistency is critical. We define our network settings in a single file called `cluster_env.sh` so we can easily source it in future labs.

```
cat << 'EOF' > cluster_env.sh
#!/bin/bash
export SMS_NAME="master"
export SMS_IP="10.0.0.1"
export SMS_ETH_INTERNAL="bond0"
export COMPUTE_ETH_INTERNAL="eth1"
export INTERNAL_NETMASK="255.255.255.0"
export INTERNAL_NETWORK="10.0.0.0"
export IPV4_GATEWAY="10.0.0.1"
export DNS_SERVERS="8.8.8.8"
export NTP_SERVER="pool.ntp.org"
export C_IP[0]="10.0.0.101"
export C_MAC[0]="52:54:00:22:33"
export C_NAME[0]="compute-01"
```

```
export C_IP[1]="10.0.0.102"
export C_MAC[1]="52:54:00:44:55:66"
export C_NAME[1]="compute-02"
echo "Cluster environment loaded."
EOF
source cluster_env.sh
```

4. Preparing Empty Compute Nodes

Unlike standard virtual machines, our compute nodes will be provisioned dynamically over the network (PXE) by the master node using Warewulf in Week 3. We **do not** install an OS on them!

1. Click **Create VM** in Proxmox. Set the Name to `compute-01`
2. **OS:** Select **Do not use any media**
3. **System & Disks:** Leave defaults (allocating 20 GB).
4. **CPU & Memory:** Allocate 1 Core and 1024 MB (1 GB) RAM.
5. **Network:** Select the internal cluster bridge (e.g., `vmbr1`) instead of the internet bridge. Finish creation.
6. Repeat the process to create `compute-02`.

5. Retrieving MAC Addresses

To provision these nodes later, the Master Node needs to know their exact MAC addresses.

1. Select `compute-01` in Proxmox and go to **Hardware**.

2. Click on **Network Device** to view its details.
3. Copy the **MAC address**.
4. Log into your `master` node and edit your `cluster_env.sh` file. Replace the placeholder `C_MAC[0]` with this real MAC address.
5. Repeat for `compute-02` and update `C_MAC[1]`.
6. Run `source cluster_env.sh` again to update the variables in memory.

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6. Submission

We are now fully prepared for Week 3.

Submit a screenshot showing:

1. Your Proxmox GUI with the `master`, `compute-01`, and `compute-02` VMs created.
2. The terminal of your Master Node showing the output of `cat cluster_env.sh` with the real MAC addresses updated.

Save this screenshot as `Week2_Lab_Surname.png`